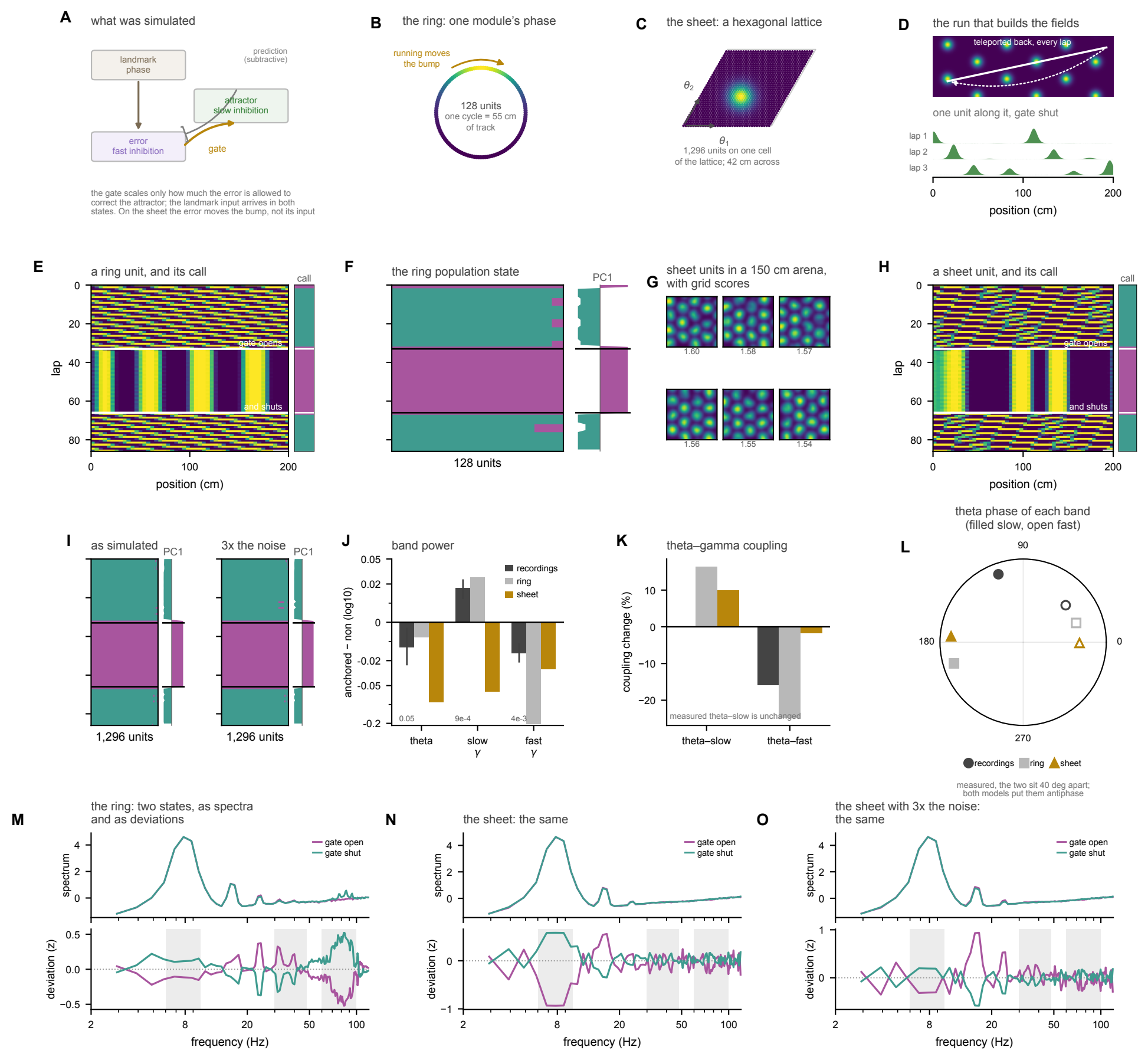

A
what was simulated
landmark phase
prediction (subtractive)
attractor slow inhibition
error fast inhibition
gate
the gate scales only how much the error is allowed to correct the attractor; the landmark input arrives in both states. On the sheet the error moves the bump, not its input
B
the ring: one module's phase
running moves the bump
128 units
one cycle = 55 cm of track
C
the sheet: a hexagonal lattice
θ2
θ1
1,296 units on one cell of the lattice; 42 cm across
D
the run that builds the fields
teleported back, every lap
one unit along it, gate shut
lap 1
lap 2
lap 3
0
100
200
position (cm)
E
a ring unit, and its call
call
lap
gate opens
and shuts
position (cm)
F
the ring population state
PC1
128 units
G
sheet units in a 150 cm arena, with grid scores
1.60
1.58
1.57
1.56
1.55
1.54
H
a sheet unit, and its call
call
gate opens
and shuts
position (cm)
I
as simulated
PC1
3x the noise
PC1
1,296 units
1,296 units
J
band power
recordings
ring
sheet
anchored − non (log10)
0.05
9e-4
4e-3
theta
slow γ
fast γ
K
theta–gamma coupling
coupling change (%)
measured theta–slow is unchanged
theta–slow
theta–fast
L
theta phase of each band (filled slow, open fast)
90
180
0
270
recordings
ring
sheet
measured, the two sit 40 deg apart; both models put them antiphase
M
the ring: two states, as spectra and as deviations
gate open
gate shut
spectrum
deviation (z)
frequency (Hz)
N
the sheet: the same
gate open
gate shut
spectrum
deviation (z)
frequency (Hz)
O
the sheet with 3x the noise: the same
gate open
gate shut
spectrum
deviation (z)
frequency (Hz)